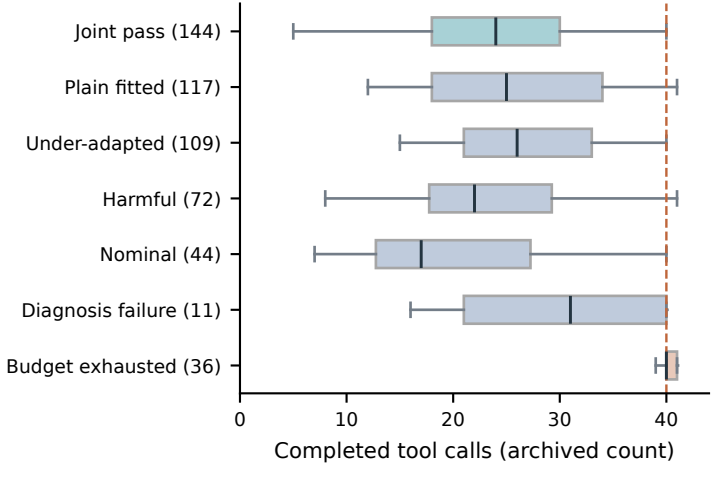

B Trace-grounded failure paths

Prediction/label shapes differ	> Broadcasted validation loss	> Harmful decoder
Class index used as physical ID	> Wrong regime dispatch	> Insufficient repair
49 qubits reported; 21 changed	> Localization Jaccard < 0.5	> Good repair; wrong diagnosis

Selected cases; arrows summarize observed code paths, not controlled causal effects or measured prevalence.